University of Tabuk
Faculty of Computers & Information Technology
Department of Computer Engineering

MICROPROCESSORS & EMBEDDED SYSTEMS LAB
CEN-412

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Project (6). Arduino Time Attendance System with RFID 20 points

1. Project description:

This project implements an RFID-based Time and Attendance System using an Arduino microcontroller, an MFRC522 RFID reader, and a Real-Time Clock (RTC) module.
The system automatically records employee check-ins and check-outs by scanning RFID tags. Each scanned tag generates a unique User ID (UID) which is then logged with the exact date and time.

The main purpose of this system is to replace traditional manual attendance taking with an automated, accurate, and efficient digital solution. The project improves reliability, reduces human error, and provides real-time attendance tracking.

Key features of the system include:

Reading RFID cards using the MFRC522 module

Automatic date and time stamping using the DS1307/DS3231 RTC

Sending attendance records to the computer via serial communication

Saving attendance data into a database system (Microsoft Access)

Differentiating between "On-time" and "Late" check-ins based on a predefined schedule

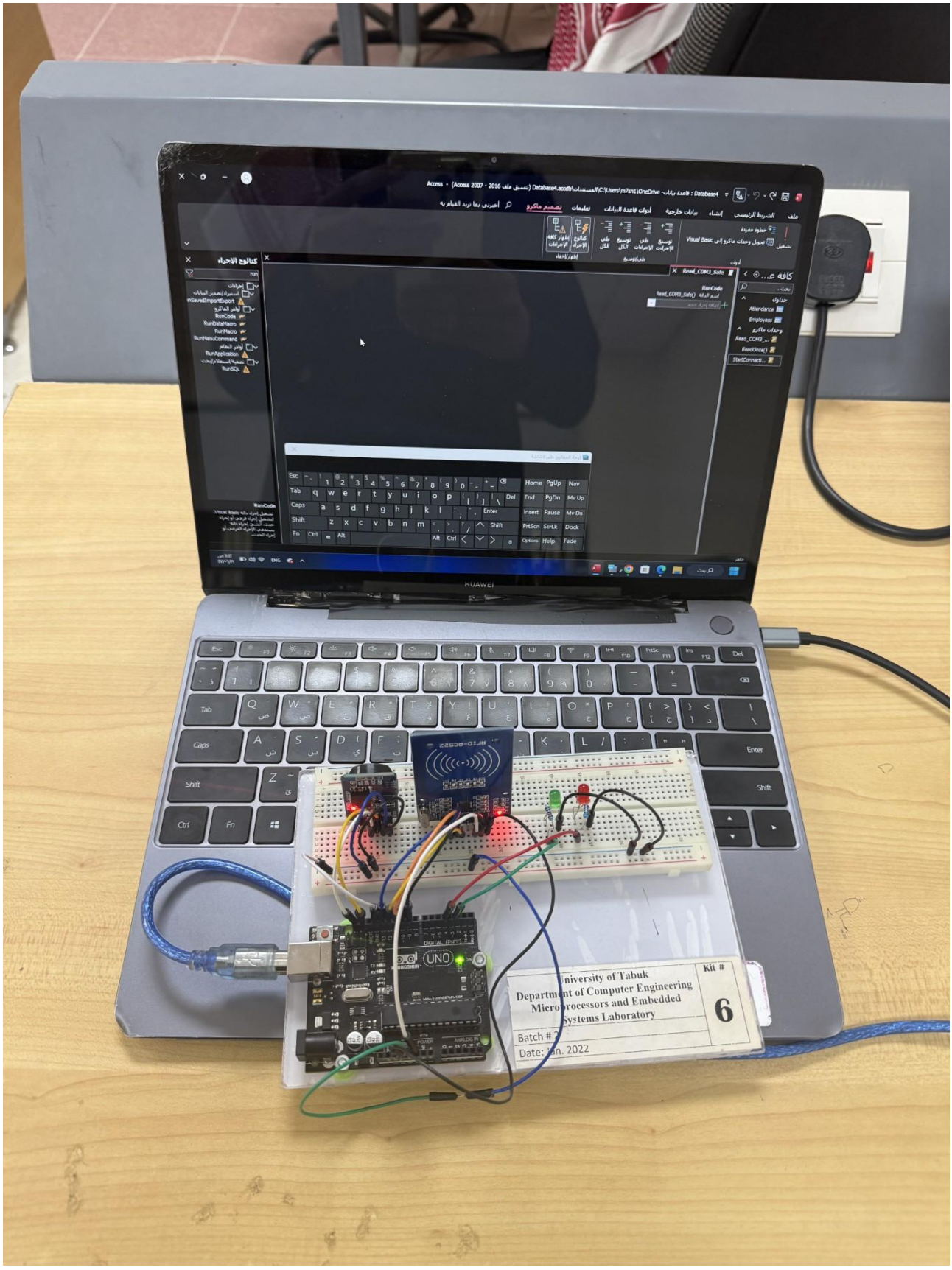
Supporting check-in and check-out logging

Reducing administrative workload and enabling automated attendance reports.

This system is suitable for small offices, student classrooms, workshops, and any environment requiring automated access logging.

2. Project schematic Diagram

In lab:



3. Algorithm and Program source code:

Algorithm Steps :

Step 1 – System Initialization
- Initialize SPI communication.
- Initialize the MFRC522 RFID reader.
- Initialize the RTC module.
- Set the check-in reference time (e.g., 08:00 AM).

Step 2 – Continuous Tag Scanning
- The Arduino continuously monitors for new RFID cards near the reader.
- If no card is detected, the loop repeats.

Step 3 – Read RFID Tag UID
- When a card is detected, read the UID bytes.
- Convert UID to a readable string.

Step 4 – Get Current Date and Time
- Read real-time data from the RTC.
- Format the date and time properly.

Step 5 – Determine Attendance Status
- Compare the current time with the official check-in time.
- If time ≤ 08:00 → Status = ON-TIME
- Else → Status = LATE

Step 6 – Send Data to the Computer
- Send the following string via Serial communication:

UID:XXXXXX | Name:EmployeeName | Date:DD/MM/YYYY | Time:HH:MM:SS | Status:IN

Step 7 – Save Record in Access Database
- Access reads the serial data and inserts it into the Attendance table.

Program Source Code:

```
#include <SPI.h>
#include <MFRC522.h>
#include <RTClib.h>

#define SS_PIN 10
#define RST_PIN 9

MFRC522 rfid(SS_PIN, RST_PIN);
RTC_DS1307 rtc;

String employee = "Abdulmohsen";

void setup() {
  Serial.begin(9600);
  SPI.begin();
  rfid.PCD_Init();
  rtc.begin();
}

void loop() {
  if (!rfid.PICC_IsNewCardPresent() || !rfid.PICC_ReadCardSerial())
    return;

  DateTime now = rtc.now();

  char dateStr[20];
  sprintf(dateStr, "%02d/%02d/%04d", now.day(), now.month(), now.year());

  char timeStr[20];
  sprintf(timeStr, "%02d:%02d:%02d", now.hour(), now.minute(), now.second());

  String UID = "";
  for (byte i = 0; i < rfid.uid.size; i++)
    UID += String(rfid.uid.uidByte[i], HEX);
  UID.toUpperCase();

  Serial.print("UID:");
  Serial.print(UID);
  Serial.print(" | Name:");
  Serial.print(employee);
  Serial.print(" | Date:");
  Serial.print(dateStr);
  Serial.print(" | Time:");
  Serial.println(timeStr);

  delay(1000);
}
```

4. Program description:

The program begins by initializing all connected components: the RFID reader, the RTC clock, and the communication interface with the computer. Once powered, the Arduino enters an infinite loop where it continuously checks for new RFID tags.

When a tag is placed near the MFRC522 reader, the Arduino retrieves its unique UID. At the same moment, the system fetches the current date and time from the RTC. These values are formatted into a readable data packet and sent to the computer via the Serial port.

Microsoft Access (or another database system) listens to these serial messages and records them in a structured table. Each entry contains:

- UID of the RFID card
- Employee name
- Date of reading
- Time of reading
- Attendance status (IN/OUT or On-time/Late depending on logic)

This automation eliminates manual attendance tracking and ensures accurate, timestamped logging of all employee entries and exits.

The program is modular, easy to modify, and reliable for real-time attendance management.